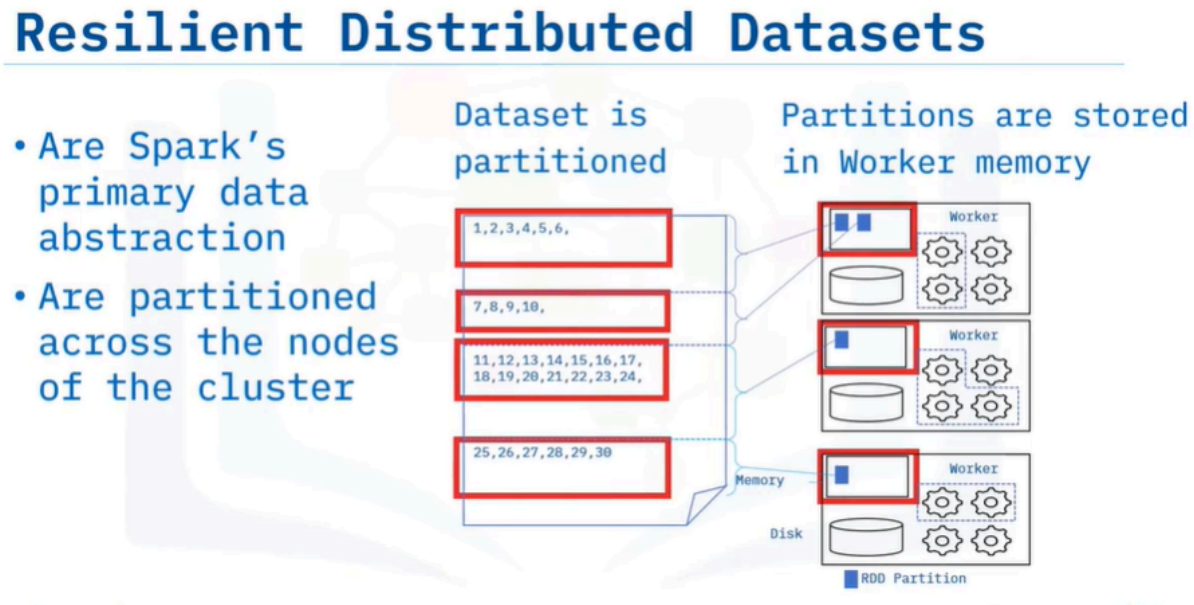
RDDs in Parallel Programming and Spark

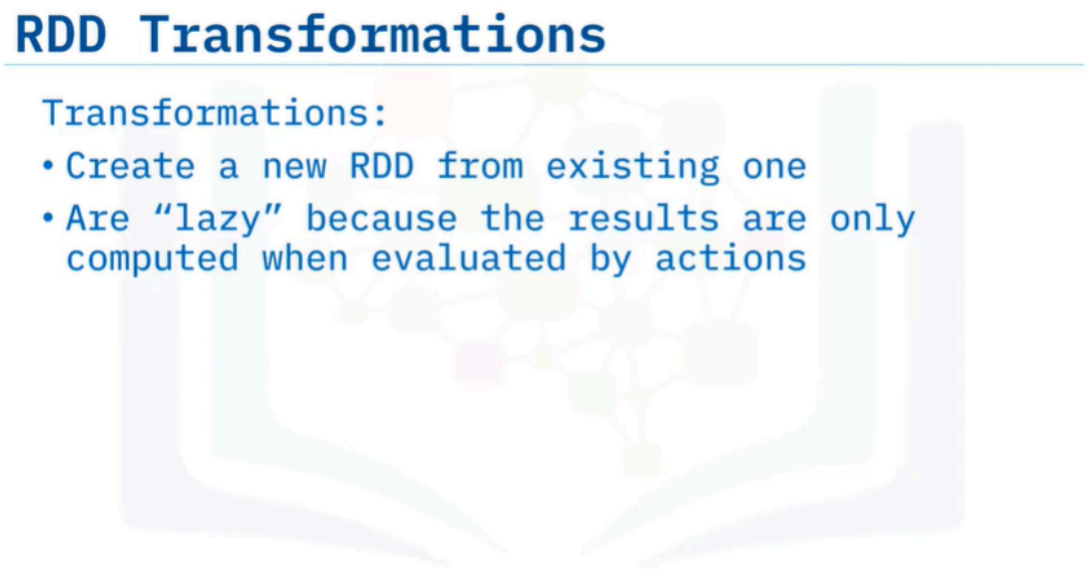
Sure! Let's talk about Resilient Distributed Datasets (RDDs) in Spark, which is a key concept in parallel programming.

Imagine you have a big library filled with books, and you want to find specific information quickly. Instead of searching through every book one by one, you can divide the books among several friends, and each friend looks through their assigned books. This is similar to how RDDs work in Spark. RDDs are collections of data that are split across different computers (or nodes) in a cluster, allowing for faster processing. When you want to change or analyze this data, you can use transformations, which create new RDDs based on the original ones, and actions, which actually perform the computations and return results.

For example, if you want to add 1 to every number in a list, you would use a transformation called "map." Each number is processed individually, and a new list is created. But the actual work only happens when you call an action, like "collect," which gathers all the results back to you. This way, Spark can efficiently manage large datasets while ensuring that if something goes wrong, it can recover easily.

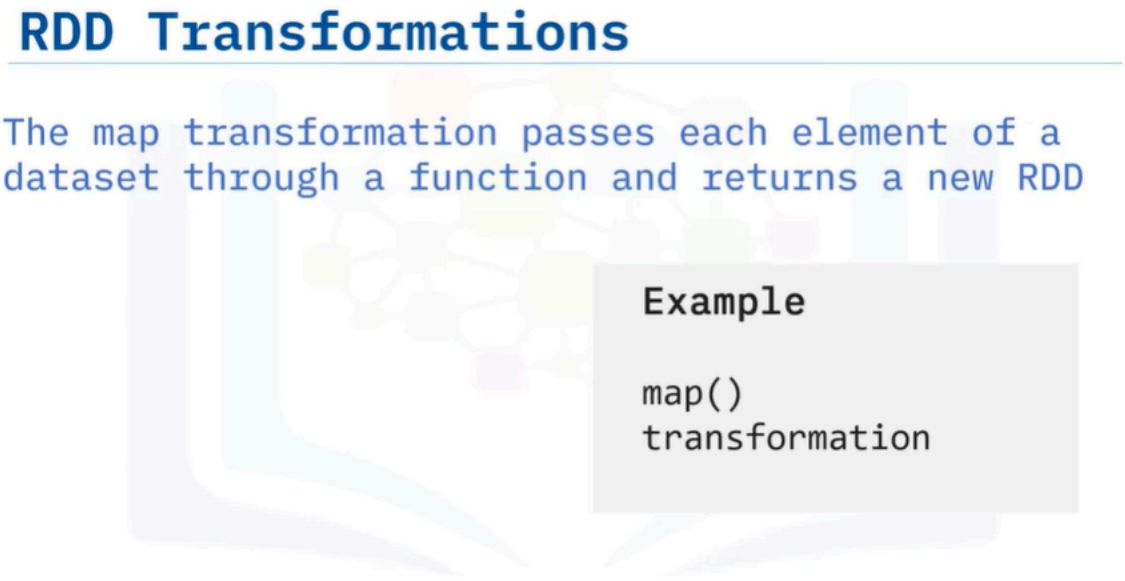


- Resilient Distributed Datasets, known as RDDs, are Spark's primary data abstraction and are partitioned across a cluster's nodes.



- The first RDD operation to explore is a Transformation. An RDD transformation creates a new RDD from an existing RDD. Transformations in Spark are

considered lazy because Spark does not compute transformation results immediately. Instead, the results are only computed when evaluated by "actions."



- For example, the map transformation passes each element of a dataset through a function resulting in a new RDD.

RDD Actions

Actions return a value to driver program after running a computation

Example

reduce()

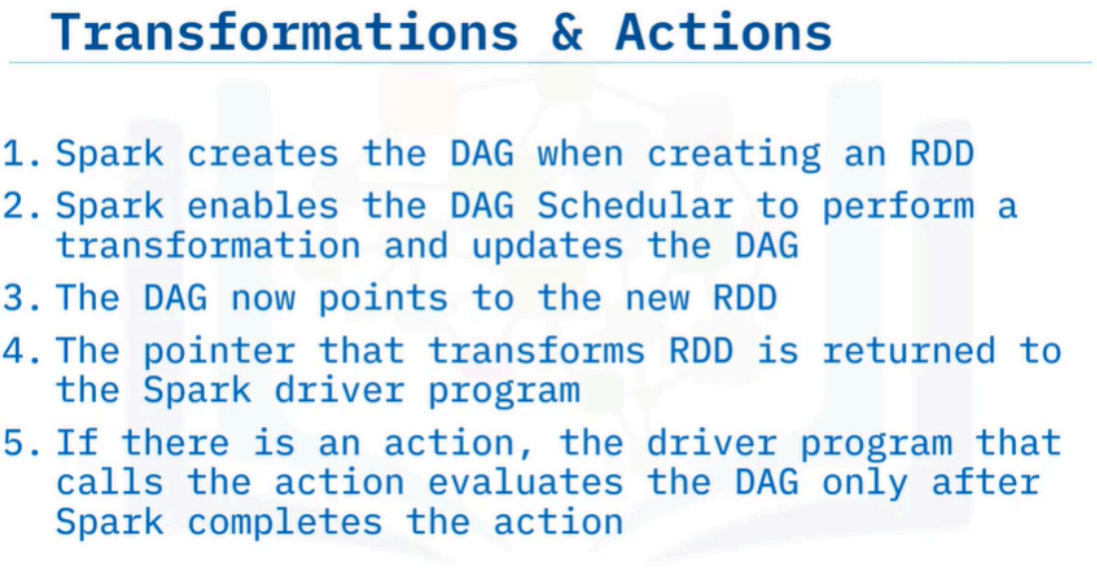
An action that aggregates all RDD elements

- To evaluate a transformation in Spark, you use an action. The action returns a value to the driver program after running a computation. One example is the reduce action, which aggregates all the elements of an RDD and returns the result to the driver program.

The Directed Acyclic Graph (DAG)

- A graphical data structure with edges and vertices
- Every new edge is obtained from an older vertex
- In Apache Spark DAG, the vertices represents RDDs and the edges represent operations such as transformations or actions
- If a node goes down, Spark replicates the DAG and restores the node.

- But how do transformations and actions happen? Spark uses a unique data structure called a Directed Acyclic Graph, knowns as a DAG, and an associated DAG Scheduler to perform RDD operations. Think of a DAG as a graphical structure composed of edges and vertices. The term "acyclic" means that new edges only originate from an existing vertex. In general, the vertices and edges are sequential.
- The vertices represent RDDs, and the edges represent the transformations or actions. The DAG Scheduler applies the graphical structure to run the tasks that use the RDD to perform transformation processes. So, why does Spark use DAGs? DAGS help enable fault tolerance. When a node goes down, Spark replicates the DAG and restores the node.



- Let's look at the process in more detail. First, Spark creates DAG when creating an RDD. Next, Spark enables the DAG Schedular to perform a transformation and updates the DAG. The DAG now points to the new RDD. The pointer that transforms RDD is returned to the Spark driver program. And, if there is an action, the driver program that calls the action evaluates the DAG only after Spark completes the action. Here are some examples of RDD transformations and actions.

Transformation examples

Transformation	Description
map (*func*)	Returns a new distributed dataset formed by passing each element of the source through a function *func*
filter (*func*)	Returns a new dataset formed by selecting those
distinct ([numTasks]))	Returns a new dataset that contains the distinct elements of the source dataset
flatmap (*func*)	Similar to map (*func*) Can map each input item to zero or more output items *Func* should return a Seq rather than a single item

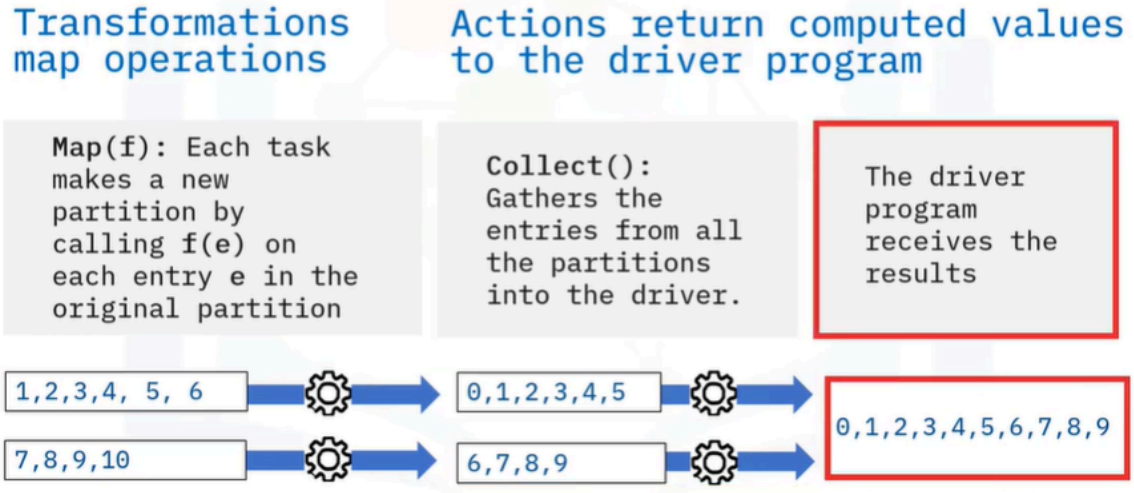
- We have already mentioned, map, which is an essential operation and capable of expressing all transformations needed in data science. Some of the more common high-level transformations making use of map and reduce include: filter, which enables filtering the elements of a dataset based on a function, and distinct, which finds the number of distinct elements in a dataset. In addition, the flatmap transformation, which is similar to the map transformation, can map each input item to zero or greater output items. Its function should return a sequence rather than a single item.

Action examples

Action	Description
reduce(*func*) aggregates dataset elements using the function *func*	*func* takes two arguments and returns one • Is commutative • Is associative • Can be computed correctly in parallel
take(*n*)	Returns an array with the first *n* element
collect()	Returns all the elements as an array WARNING: Make sure that ? will fit in driver program
takeOrdered (*n*,key=*func*)	Returns *n* elements ordered in ascending order or as specified by the optional key function

- Next, frequently used actions include: reduce, which aggregates dataset elements using the function func. take, which returns an array with the first n element, collect, which returns all the elements of the dataset as an array and takeOrdered, which returns elements ordered in ascending order or as specified by an optional function argument. You can find more details about transformations and actions, on the Spark.Apache.org website.

A Transformation with an Action



- A transformation is only a mapping of operations. You need actions to get the computed values to the driver program: This example considers a function f of x that decrements x by 1. So, f of x equals x minus 1. You'll apply this function as a transformation to a dataset using the map transformation. Each task makes a new partition by calling f of e on each entry e in the original partition. Then, the collect action gathers the entries from all the partitions into the driver, which receives the results.